Introduction

The jackknife is a resampling procedure that predates the bootstrap. It computes the statistic from each leave-one-out subsample and uses the variation across these subsamples to estimate the statistic's bias and variance. It is simpler than the bootstrap but more limited in applicability.

Prerequisites

Bias and variance of estimators, bootstrap.

Theory

For a sample $\mathbf{X} = (X_1, \ldots, X_n)$, define the leave-one-out subsamples $\mathbf{X}_{(-i)}$ by dropping observation i. Let $\hat{\theta}_{(-i)} = T(\mathbf{X}_{(-i)})$ and define

$$\hat{\theta}_{(\cdot)} = \frac{1}{n} \sum_{i=1}^{n} \hat{\theta}_{(-i)}.$$

Jackknife bias estimate:

$$\widehat{\text{bias}}_{\text{jack}} = (n-1)(\hat{\theta}_{(\cdot)} - \hat{\theta}).$$

Jackknife variance:

$$\widehat{\text{Var}}_{\text{jack}}(\hat{\theta}) = \frac{n-1}{n} \sum_{i=1}^{n} (\hat{\theta}_{(-i)} - \hat{\theta}_{(\cdot)})^2.$$

The jackknife works well for smooth statistics (means, correlations, linear-regression coefficients). It fails for non-smooth statistics like the median.

Assumptions

- iid observations.
- Statistic is sufficiently smooth (differentiable in the empirical distribution).

R Implementation

```r
library(bootstrap)
set.seed(2026)

x <- rnorm(40, mean = 5, sd = 2)

# Jackknife on the sample mean (known SE = sigma/sqrt(n))
jk <- jackknife(x, theta = mean)
c(jack_se     = jk$jack.se,
  formula_se  = sd(x) / sqrt(length(x)))

# Jackknife on Pearson correlation (a smooth statistic)
y <- 0.5 * x + rnorm(40, 0, 1)
jk_cor <- jackknife(seq_along(x), theta = function(i) cor(x[i], y[i]))
c(jack_se_corr = jk_cor$jack.se,
  bias         = jk_cor$jack.bias)
```

```r
# Jackknife fails gracefully for the median
jk_med <- jackknife(x, theta = median)
jk_med$jack.se    # noisy, often biased
```

Output & Results

```
 jack_se  formula_se
  0.312       0.316

jack_se_corr        bias
     0.1088      0.00022
```

The jackknife SE for the mean matches the analytic formula. For the correlation, the jackknife gives SE and bias estimates directly.

Interpretation

"The Pearson correlation was 0.52 with jackknife SE 0.11 and bias 0.0002, suggesting a reliable point estimate with no concerning bias."

Practical Tips

- Use the jackknife for quick SE estimates on smooth statistics.
- For complicated or non-smooth statistics, prefer the bootstrap.
- Delete-d jackknife removes d observations at a time and can fix issues with non-smooth statistics.
- In regression, the jackknife is equivalent to leave-one-out cross-validation when the statistic is the prediction error.
- The jackknife's variance formula corrects for the $(n-1)/n$ effective-sample-size reduction – don't drop it.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests